

CISCO PACKET TRACER NETWORK LABS

Internship Project Report

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CISCO PACKET TRACER NETWORK LABS

Internship Project Report

Project Repository: <https://github.com/SandeepSandy08/Cisco-Packet-Tracer-Networking-Labs.git>

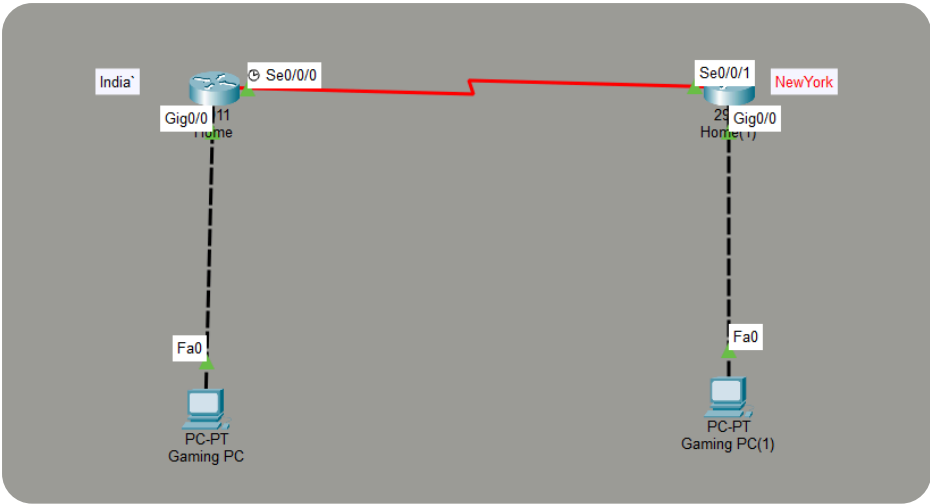
INTRODUCTION

This document contains practical networking labs performed using Cisco Packet Tracer, covering basic configuration, VLANs, routing, network services, security, and troubleshooting. Each task includes its configuration and verification results.

BASIC CISCO ROUTER CONFIGURATION

To understand how two PCs in different locations communicate over a network by configuring two routers connected through a Wide Area Network (WAN) connection. This task focuses on establishing long-distance routing between separate local networks.

NETWORK TOPOLOGY:



DEVICES USED:

- 1. 2911 Router - 2
- 2. PC(s) - 2
- 3. Cable type - Cable Cross-over , Serial DCE
- 4. Module Installed - HWIC-T2 for Both Router

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
Router 1 LAN	Gig0/0	192.168.11.1	255.255.255.0
Router 1 WAN	Se0/0/0	192.168.12.1	255.255.255.0
PC 1	NIC - Fa0	192.168.11.8	255.255.255.0
Router 2 LAN	Gig0/0	192.168.13.1	255.255.255.0
Router 2 WAN	Se0/0/1	192.168.12.2	255.255.255.0
PC 2	NIC - Fa0	192.168.13.3	255.255.255.0

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CONFIGURATION :

- PC - Assign distinct static IP addresses and default gateways on Both PCs
- Router - Connect Both Router together. Set up the local network port on each router, and manually type the path to the other router's network.
- The GitHub Repository Contains :
 1. The router and PC configurations were implemented in Cisco Packet Tracer.
 2. Configuration files: Available in the GitHub repository under the Config Files folder.
 3. Packet Tracer lab file: Available under the Labs folder.

VERIFICATION:

In Router CLI (type) - show ip interface brief or sh ip int br

To Verify

- Router interfaces.
- Verify IP addressing.
- Test connectivity between PC0 and PC1 using ping.
- Confirm successful communication through the WAN connection

RESULT

The PCs were successfully connected through the two routers, and ping verification confirmed successful communication across the WAN connection.

```
C:\>ping 192.168.13.3

Pinging 192.168.13.3 with 32 bytes of data:

Reply from 192.168.13.3: bytes=32 time=27ms TTL=126
Reply from 192.168.13.3: bytes=32 time=20ms TTL=126
Reply from 192.168.13.3: bytes=32 time=2ms TTL=126
Reply from 192.168.13.3: bytes=32 time=19ms TTL=126

Ping statistics for 192.168.13.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 27ms, Average = 17ms
```

Ping From PC1 to PC2

```
C:\>ping 192.168.11.8

Pinging 192.168.11.8 with 32 bytes of data:

Reply from 192.168.11.8: bytes=32 time=34ms TTL=126
Reply from 192.168.11.8: bytes=32 time=17ms TTL=126
Reply from 192.168.11.8: bytes=32 time=16ms TTL=126
Reply from 192.168.11.8: bytes=32 time=37ms TTL=126

Ping statistics for 192.168.11.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 37ms, Average = 26ms
```

Ping From PC2 to PC1

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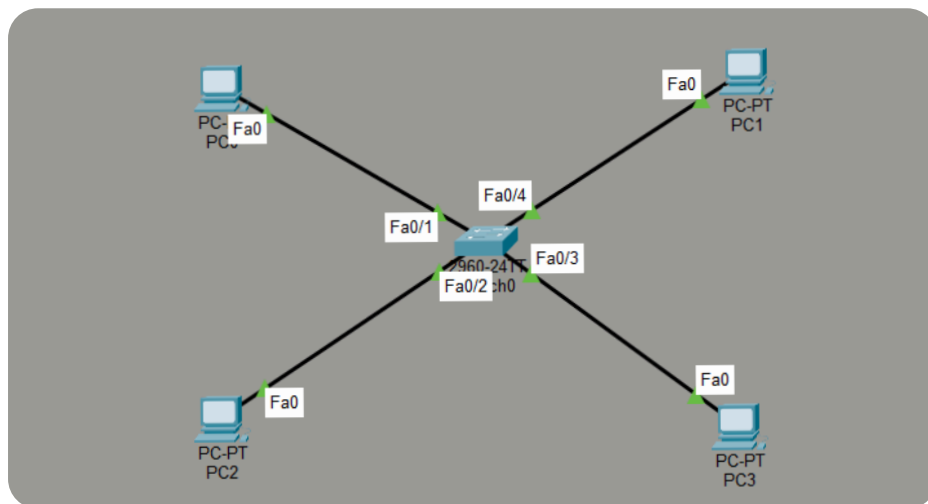
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BASIC CISCO SWITCH CONFIGURATION

To understand how to connect multiple PCs and enable them to communicate across a network by configuring switches connected through a LAN. This task focuses on utilizing a central switch to interconnect multiple PCs and establish basic local area network communication.

NETWORK TOPOLOGY:



DEVICES USED:

1. 2960 Switch - 1
2. PC(s) - 4
3. Cable type - Straight-Through Cable

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
	Gateway	192.168.10.1	255.255.255.0
PC 0	NIC - Fa0 - Fa0/1	192.168.10.2	255.255.255.0
PC 1	NIC - Fa0 - Fa0/2	192.168.10.3	255.255.255.0
PC 2	NIC - Fa0 - Fa0/3	192.168.10.4	255.255.255.0
PC 3	NIC - Fa0 - Fa0/4	192.168.10.5	255.255.255.0

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VERIFICATION:

In Switch CLI (type) - show ip interface brief or sh ip int br

To Verify

- Switch interfaces.
- Verify IP addressing.
- Test connectivity between PC to PC using ping.

RESULT

The PCs were successfully connected through the switch and verified using ping, confirming basic local network communication.

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=16ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=6ms TTL=128
Reply from 192.168.10.2: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 7ms
```

Ping From PC1 to PC2

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping From PC1 to PC3

```
C:\>ping 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=10ms TTL=128

Ping statistics for 192.168.10.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Ping From PC1 to PC3

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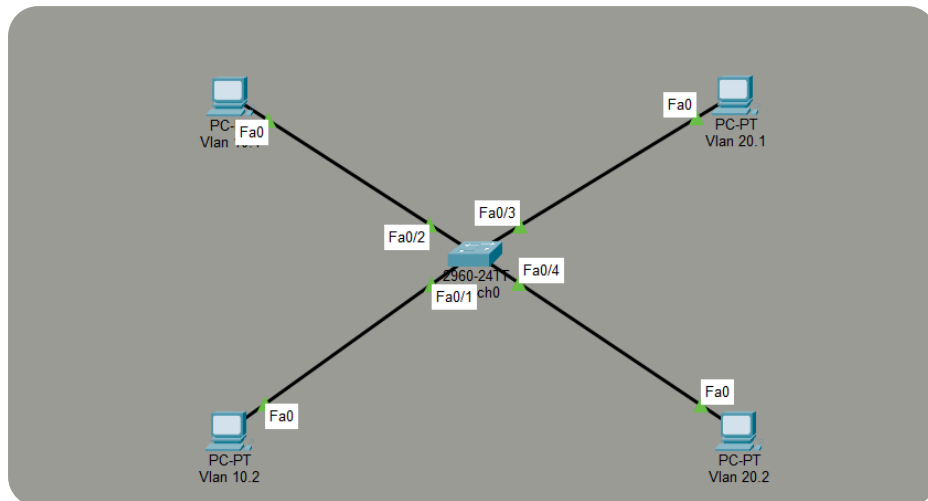
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VLAN CONFIGURATION AND VERIFICATION

To understand how to segregate a local network by configuring and verifying Virtual Local Area Networks (VLANs) on a switch to control broadcast domains and secure communication.

NETWORK TOPOLOGY:



DEVICES USED:

1. 2960 Switch - 1
2. PC(s) - 4
3. Cable type - Straight-ThroughCable

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
Switch	Vlan 10 Gateway	192.168.11.8	255.255.255.0
PC 0	NIC - Fa0 - Fa0/1	192.168.11.9	255.255.255.0
PC 1	NIC - Fa0 - Fa0/2	192.168.11.10	255.255.255.0
Switch	Vlan 20 Gateway	192.168.13.3	255.255.255.0
PC 2	NIC - Fa0 - Fa0/3	192.168.13.4	255.255.255.0
PC 3	NIC - Fa0 - Fa0/4	192.168.13.5	255.255.255.0

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CONFIGURATION :

- Switch - Create VLAN 10 and VLAN 20 to segment the network (VLAN IDs range from 1 to 4094).
- Port Assignment: Assign the respective switch ports to each VLAN, restricting them to their own broadcast domains
- The GitHub Repository Contains :
 1. The router, Switch, VLAN and PC configurations were implemented in Cisco Packet Tracer.
 2. Configuration files: Available in the GitHub repository under the Config Files folder.
 3. Packet Tracer lab file: Available under the Labs folder.

VERIFICATION:

In Switch CLI (type) - show vlan brief

To Verify

- Switch interfaces.
- Verify IP addressing.
- Test connectivity between VLAN 10 and VLAN 20 using ping.

RESULT

Pings between the same VLAN succeed, while pings between different VLANs fail (packets drop). This verifies that the switch correctly isolates network traffic using VLAN

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.9

Pinging 192.168.11.9 with 32 bytes of data:

Reply from 192.168.11.9: bytes=32 time=15ms TTL=128
Reply from 192.168.11.9: bytes=32 time<1ms TTL=128
Reply from 192.168.11.9: bytes=32 time=6ms TTL=128
Reply from 192.168.11.9: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 5ms
```

Ping From VLAN 10 to VLAN 10

```
C:\>ping 192.168.13.4

Pinging 192.168.13.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.13.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100%
```

Ping From VLAN 10 to VLAN 20

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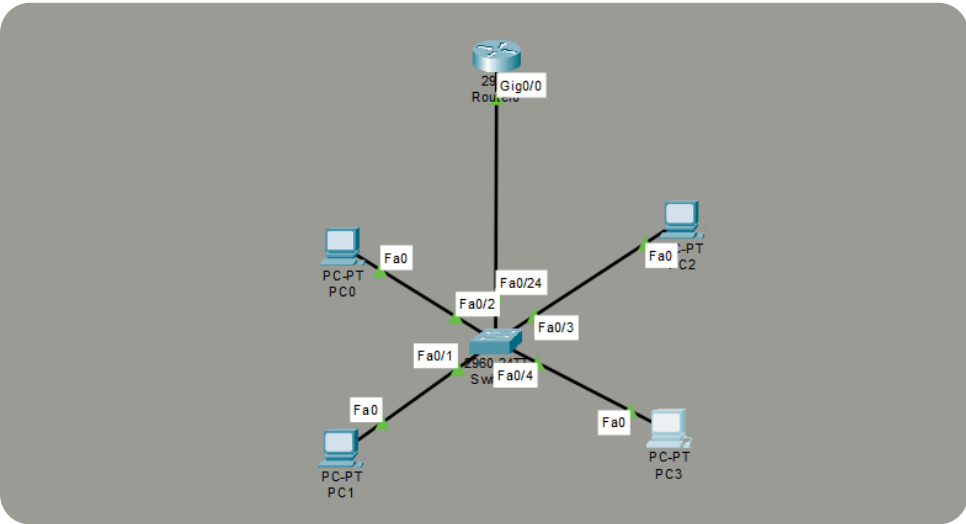
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Project Repository: <https://github.com/SandeepSandy08/Cisco-Packet-Tracer-Networking-Labs.git>

INTER-VLAN ROUTING USING ROUTER-ON-A-STICK

To understand how PCs in different VLANs communicate by configuring Router-on-a-Stick (ROAS), utilizing subinterfaces on a single router port to enable inter-VLAN routing

NETWORK TOPOLOGY:



DEVICES USED:

- 1. 2960 Switch - 1
- 2. 2901 Router 1
- 3. PC(s) - 4
- 4. Cable type - Straight-Through Cable

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
Router	Vlan 10 Gateway	192.168.11.8	255.255.255.0
PC 0	NIC - Fa0 - Fa0/1	192.168.11.9	255.255.255.0
PC 1	NIC - Fa0 - Fa0/2	192.168.11.10	255.255.255.0
Router	Vlan 20 Gateway	192.168.13.3	255.255.255.0
PC 2	NIC - Fa0 - Fa0/3	192.168.13.4	255.255.255.0
PC 3	NIC - Fa0 - Fa0/4	192.168.13.5	255.255.255.0

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CONFIGURATION :

- Switch - Create VLAN 10 and VLAN 20 to segment the network (VLAN IDs range from 1 to 4094).
- Port Assignment: Assign the respective switch ports to each VLAN, restricting them to their own broadcast domains
- The GitHub Repository Contains :
 - 1.The router, Switch, VLAN and PC configurations were implemented in Cisco Packet Tracer.
 - 2.Configuration files: Available in the GitHub repository under the Config Files folder.
 - 3.Packet Tracer lab file: Available under the Labs folder.

VERIFICATION:

In Switch CLI (type) - show vlan brief

To Verify

- Switch interfaces.
- Verify IP addressing.
- Test connectivity between VLAN 10 and VLAN 20 using ping.

RESULT

Pings between different VLANs now succeed. This confirms that the router is successfully routing traffic between VLAN 10 and VLAN 20 using virtual subinterfaces (.10 and .20).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.13.4

Pinging 192.168.13.4 with 32 bytes of data:

Reply from 192.168.13.4: bytes=32 time<1ms TTL=127
Reply from 192.168.13.4: bytes=32 time<1ms TTL=127
Reply from 192.168.13.4: bytes=32 time<1ms TTL=127
Reply from 192.168.13.4: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.13.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping From VLAN 10 to VLAN 20

```
C:\>ping 192.168.13.5

Pinging 192.168.13.5 with 32 bytes of data:

Reply from 192.168.13.5: bytes=32 time<1ms TTL=127
Reply from 192.168.13.5: bytes=32 time<1ms TTL=127
Reply from 192.168.13.5: bytes=32 time<1ms TTL=127
Reply from 192.168.13.5: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.13.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping From VLAN 10 to VLAN 20

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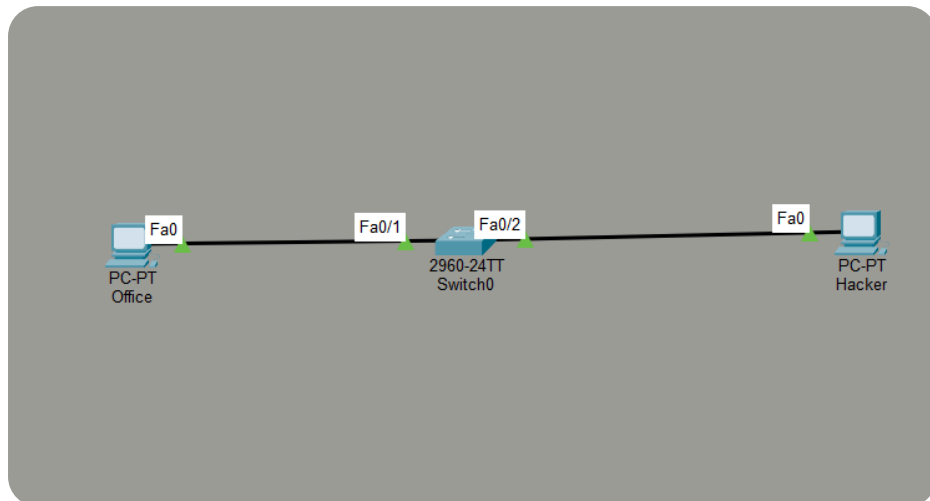
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SWITCH PORT SECURITY

To understand how to protect network ports by configuring Switch Port Security to block unauthorized devices based on their MAC addresses.

NETWORK TOPOLOGY:



DEVICES USED:

1. 2960 Switch - 1
2. PC(s) - 2
3. Cable type - Straight-Through Cable

IP ADDRESSING :

Network / Devices	Interface	IP Address	Subnet Mask
Network Series	NIL	192.168.11.0	255.255.255.0
Default Gateway	NIL	192.168.11.1	255.255.255.0
Switch	Fa0/1 and Fa0/2	NIL	NIL
PC 1 - Office PC	NIC - Fa0 - Fa0/1	192.168.11.2	255.255.255.0
PC 2 - Hacker PC	NIC - Fa0 - Fa0/2	192.168.11.3	255.255.255.0

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CONFIGURATION :

- The GitHub Repository Contains :
 - 1.The Switch, and PC configurations were implemented in Cisco Packet Tracer.
 - 2.Configuration files: Available in the GitHub repository under the Config Files folder.
 - 3.Packet Tracer lab file: Available under the Labs folder.

VERIFICATION:

In Switch CLI (type) - show port-security interface fa0/1

To Verify

- Switch interfaces.
- Test Unauthorized Access: Connect an Hacker PC (a different MAC address) into the secured port.
- Verify Port Violation State: Check if the switch port status automatically transitions to Secure-shutdown (the port light turns red).

RESULT

- Authorized Device: The designated PC communicates successfully across the network with no packet drops.
- Unauthorized Device: Pings from a rogue device fail completely as the switch port instantly transitions into a Secure-shutdown state, successfully blocking unauthorized access.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.4

Pinging 192.168.11.4 with 32 bytes of data:

Reply from 192.168.11.4: bytes=32 time<1ms TTL=128
Reply from 192.168.11.4: bytes=32 time<1ms TTL=128
Reply from 192.168.11.4: bytes=32 time<1ms TTL=128
Reply from 192.168.11.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping From Authorized Device

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168,
Ping request could not find host 192.168,. Please check the na
C:\>ping 192.168.11.4

Pinging 192.168.11.4 with 32 bytes of data:

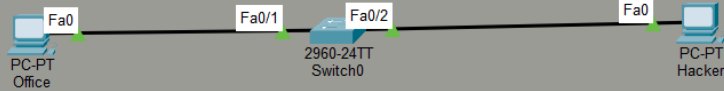
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.11.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

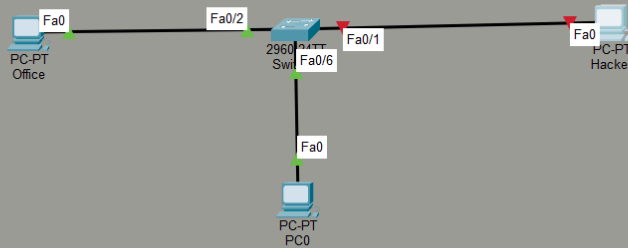
Ping From UnAuthorized Device

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Office PC is Connect to port Fa0/1
and the port is secured



when I Connect with Hacker PC to
port Fa0/1 the switch port goes into
shutdown

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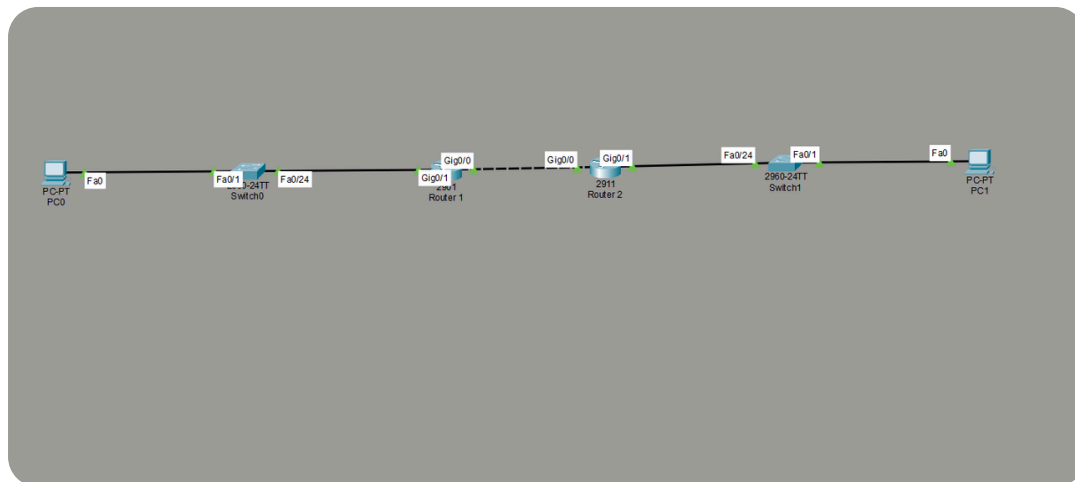
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STATIC ROUTING BETWEEN TWO ROUTERS

To establish communication between two remote PCs by manually configuring static routes on two routers connected over a WAN link.

NETWORK TOPOLOGY:



DEVICES USED:

1. 2960 Switch - 2
2. 2901 Router 1
3. 2911 Router 1
4. PC(s) - 2
5. Cable type - Straight-Through Cable, Cross-Over

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
Router 1 LAN	Gig0/1 - Fa0/24	192.168.11.1	255.255.255.0
Router 1 WAN	Gig0/0	10.0.0.1	255.255.255.0
PC 1	NIC - Fa0	192.168.11.8	255.255.255.0
Router 2 LAN	Gig0/1 - Fa0/24	192.168.13.1	255.255.255.0
Router 2 WAN	Gig0/0	10.0.0.2	255.255.255.0
PC 2	NIC - Fa0	192.168.13.3	255.255.255.0

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CONFIGURATION :

- Switch - Connect the PCs to local **Switches (Switch0 and Switch1)** to local network traffic.
 - PC - Assign distinct static IP addresses and default gateways on Both **PCs**
 - Router - Connect Both Router together. Set up the local network port on each router, and manually type the path to the other router's network.
-
- The GitHub Repository Contains :
 1. The Switch, and PC configurations were implemented in Cisco Packet Tracer.
 2. Configuration files: Available in the GitHub repository under the Config Files folder.
 3. Packet Tracer lab file: Available under the Labs folder.

VERIFICATION:

In Router CLI (type) - show ip route and show ip int br

To Verify

- Verify the static entries are active
- The routing table.
- Test connectivity by executing a ping in Pcs Cmd Line.

RESULT

- Pings between the two PCs in different locations succeed, confirming that traffic moves through the local switches and across the WAN-connected routers.

```
C:\>ping 192.168.13.3

Pinging 192.168.13.3 with 32 bytes of data:

Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.13.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping From PC1 to PC2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.8

Pinging 192.168.11.8 with 32 bytes of data:

Reply from 192.168.11.8: bytes=32 time<1ms TTL=126
Reply from 192.168.11.8: bytes=32 time=1ms TTL=126
Reply from 192.168.11.8: bytes=32 time<1ms TTL=126
Reply from 192.168.11.8: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.11.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping From PC2 to PC1

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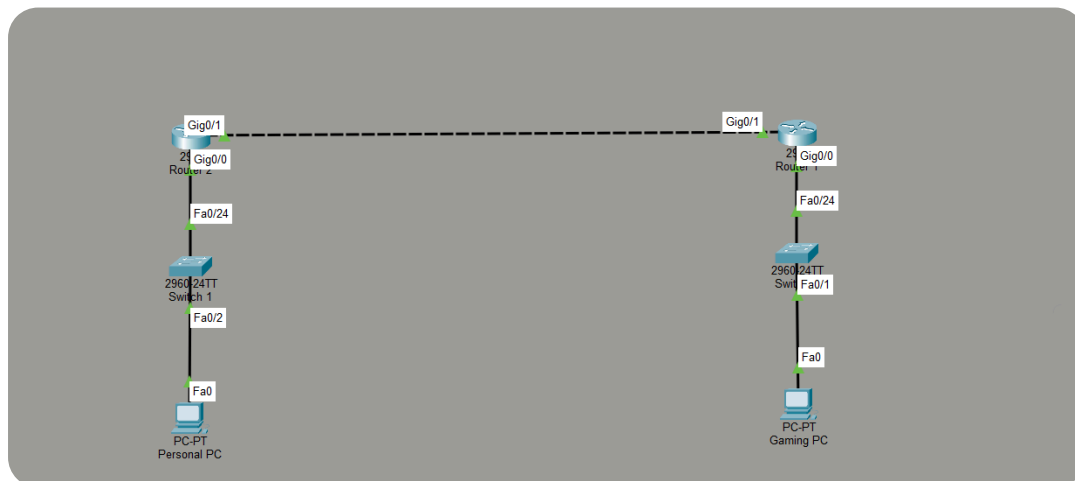
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DEFAULT ROUTE CONFIGURATION

To forward all unknown external network traffic to a next-hop gateway using a default route (0.0.0.0/0) when no specific path exists in the routing table.

NETWORK TOPOLOGY:



DEVICES USED:

1. 2960 Switch - 2
2. 2911 Router - 2
3. PC(s) - 2
4. Cable type - Straight-Through Cable, Cross-Over

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
Router 1 LAN	Gig0/0 - Fa0/24	192.168.11.1	255.255.255.0
Router 1 WAN	Gig0/1	10.0.0.1	255.255.255.0
PC 1	NIC - Fa0	192.168.11.8	255.255.255.0
Router 2 LAN	Gig0/0 - Fa0/24	192.168.13.1	255.255.255.0
Router 2 WAN	Gig0/1	10.0.0.2	255.255.255.0
PC 2	NIC - Fa0	192.168.13.3	255.255.255.0

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CONFIGURATION :

- PCs & Switches - PC0 and PC1 connect through Switch0 and Switch1, using their local router ports as default gateways.
- Router - Instead of typing specific paths for every network, configure a catch-all route (0.0.0.0) on Router 1 to send all external traffic over the cable to Router 2 (10.0.0.2), and do the same on Router 2.
- The GitHub Repository Contains :
 1. The Switch, and PC configurations were implemented in Cisco Packet Tracer.
 2. Configuration files: Available in the GitHub repository under the Config Files folder.
 3. Packet Tracer lab file: Available under the Labs folder.

VERIFICATION:

In Router CLI (type) - show ip route and show ip int br

To Verify

- The default route entry is active (marked with a S* symbol).
- Test connectivity by executing a ping in Pcs Cmd Line.

RESULT

- Pings to the remote network succeed, proving the routers successfully use the default route to send unknown traffic across the network link.

```
C:\>ping 192.168.13.3

Pinging 192.168.13.3 with 32 bytes of data:

Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.13.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping From PC1 to PC2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.8

Pinging 192.168.11.8 with 32 bytes of data:

Reply from 192.168.11.8: bytes=32 time<1ms TTL=126
Reply from 192.168.11.8: bytes=32 time=1ms TTL=126
Reply from 192.168.11.8: bytes=32 time<1ms TTL=126
Reply from 192.168.11.8: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.11.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping From PC2 to PC1

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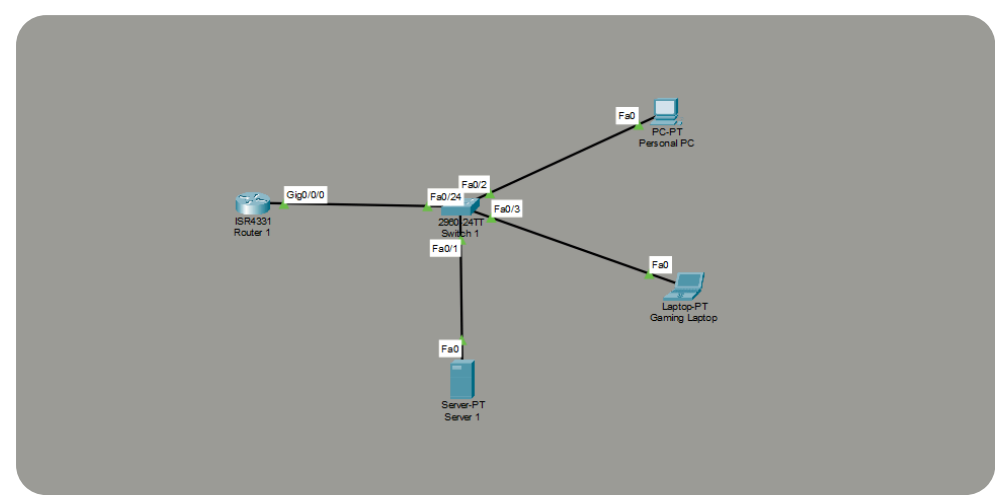
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Project Repository: <https://github.com/SandeepSandy08/Cisco-Packet-Tracer-Networking-Labs.git>

DHCP SERVER CONFIGURATION ON CISCO ROUTER

To automatically assign IP addresses, subnet masks, and default gateways to local end devices using the ISR4331 Router as a DHCP Server.

NETWORK TOPOLOGY:



DEVICES USED:

- 1. 2960 Switch - 1
- 2. 4331 Router - 1
- 3. Server 1
- 4. PC(s) - 1
- 5. Laptop - 1
- 6. Cable type - Straight-Through Cable

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
Router 1 LAN	Gig0/0/0 - Fa0/24	192.168.13.1	255.255.255.0
Switch	Vlan	NIL	NIL
Server	NIC - Fa0	192.168.13.2 (Static)	255.255.255.0
PC	NIC - Fa0	Automatic (DHCP)	255.255.255.0
Laptop	NIC - Fa0	Automatic (DHCP)	255.255.255.0

CISCO PACKET TRACER NETWORK LABS

Internship Project Report

CONFIGURATION :

- Server - Configured with a static IP address to prevent it from automatically changing.
 - PC & Laptop - Set to Automatic (DHCP) mode to receive network settings dynamically.
 - Router (DHCP Server) - Configured with a network pool to hand out IP addresses automatically. The router and server IPs are excluded from this pool so they do not accidentally change.
-
- The GitHub Repository Contains :
 1. The Switch, and PC configurations were implemented in Cisco Packet Tracer.
 2. Configuration files: Available in the GitHub repository under the Config Files folder.
 3. Packet Tracer lab file: Available under the Labs folder.

VERIFICATION:

In Router CLI (type) - show ip dhcp binding and show ip int br

To Verify

- Active automatic assignments.
- PC and Laptop display an address within the correct network range.

RESULT

- Both the Personal PC and Gaming Laptop successfully pull unique IP parameters automatically, showing "DHCP request successful" with no packet loss.

```
C:\>ping 192.168.13.3

Pinging 192.168.13.3 with 32 bytes of data:

Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126
Reply from 192.168.13.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.13.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping From Laptop to PC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.13.2

Pinging 192.168.13.2 with 32 bytes of data:

Reply from 192.168.13.2: bytes=32 time<1ms TTL=128
Reply from 192.168.13.2: bytes=32 time<1ms TTL=128
Reply from 192.168.13.2: bytes=32 time<1ms TTL=128
Reply from 192.168.13.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.13.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping From PC to Server

CISCO PACKET TRACER NETWORK LABS

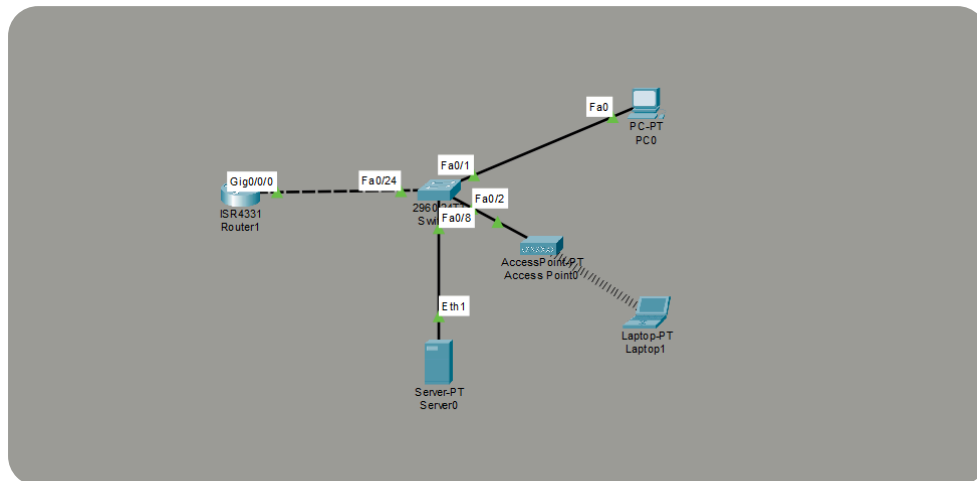
Internship Project Report

Project Repository: <https://github.com/SandeepSandy08/Cisco-Packet-Tracer-Networking-Labs.git>

DNS CONFIGURATION IN A CISCO NETWORK LAB

To understand how a Domain Name System (DNS) server translates easy-to-read domain names into network IP addresses so devices can open websites using names instead of numbers.

NETWORK TOPOLOGY:



DEVICES USED:

1. 2960 Switch - 1
2. 4331 Router - 1
3. Server 1
4. PC(s) - 1
5. Laptop - 1
6. Wired Connections / Wireless Access : Access point
7. Cable type - Straight-Through Cable, Cable Cross-Over

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
Router 1	Gig0/0/0 - Fa0/24	192.168.11.1	255.255.255.0
Switch	Vlan	NIL	NIL
Server	Eth 1 - Fa0/8	192.168.11.2 (Static)	255.255.255.0
PC	NIC - Fa0	Automatic (DHCP)	255.255.255.0
Access Point	Fa0/2	NIL	NIL
Laptop	Wireless	Automatic (DHCP)	255.255.255.0

For learning and testing purposes, I Used wireless Wi-Fi via a local Access Point to connect the laptop. However, standard wired Ethernet cables can be used instead of wireless connections, and the DNS name resolution will function exactly the same way across the network infrastructure.

CISCO PACKET TRACER NETWORK LABS

Internship Project Report

CONFIGURATION :

- Router - Set up the gateway IP, turn on the DHCP pool, and add the Server IP as the DNS server.
 - Server -Assign a permanent static IP, turn ON the Web (HTTP) and DNS services, and add your website name.
 - Cables: Plug the router, server, and Access Point into the switch using regular network cables.
 - Wireless: Leave the Access Point and Laptop on their default, unencrypted Wi-Fi settings so they link up automatically.
 - PC - Turn on Automatic (DHCP) mode on the PC and Laptop to receive network settings instantly.
-
- The GitHub Repository Contains :
 1. The Switch, and PC configurations were implemented in Cisco Packet Tracer.
 2. Configuration files: Available in the GitHub repository under the Config Files folder.
 3. Packet Tracer lab file: Available under the Labs folder.

VERIFICATION:

In Router CLI (type) — show ip dhcp binding and show ip int br
To Verify

- Active automatic assignments.
- PC and Laptop display an address within the correct network range.
- The interface state reads Up/Up on both devices, confirming the link backbone.
- Test connectivity between the end devices and the DNS server using ping.

RESULT

- The simulated website page loads perfectly onto the browser screen using only the text name entry, proving that the DNS server is properly translating domain names for both wired and wireless users across the switch architecture.

```
Router#
Router#show ip dhcp binding
^
% Invalid input detected at '^' marker.

Router#show ip dhcp binding

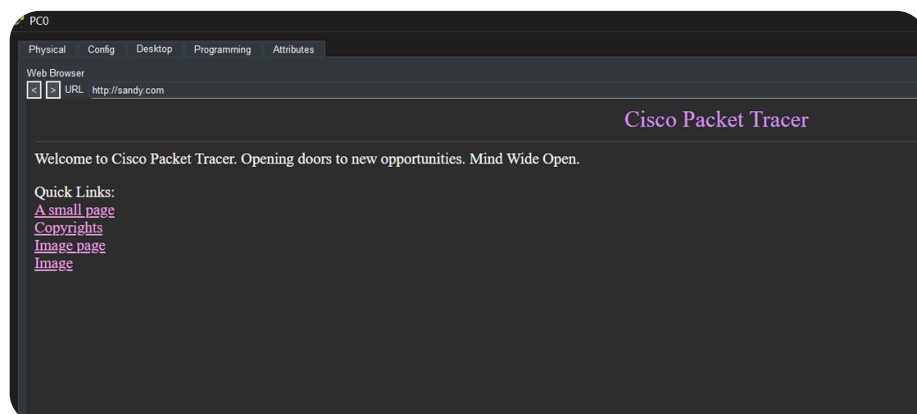

| IP address    | Client-ID/<br>Hardware address | Lease expiration | Type      |
|---------------|--------------------------------|------------------|-----------|
| 192.168.11.11 | 0002.4A01.81D8                 | --               | Automatic |
| 192.168.11.12 | 0002.4A2E.C743                 | --               | Automatic |


Router#sh ip int br


| Interface            | IP-Address   | OK? | Method | Status                |
|----------------------|--------------|-----|--------|-----------------------|
| GigabitEthernet0/0/0 | 192.168.11.1 | YES | manual | up                    |
| GigabitEthernet0/0/1 | unassigned   | YES | unset  | administratively down |
| GigabitEthernet0/0/2 | unassigned   | YES | unset  | administratively down |


Router#
Router#
```

Result of DNS Server



CISCO PACKET TRACER NETWORK LABS

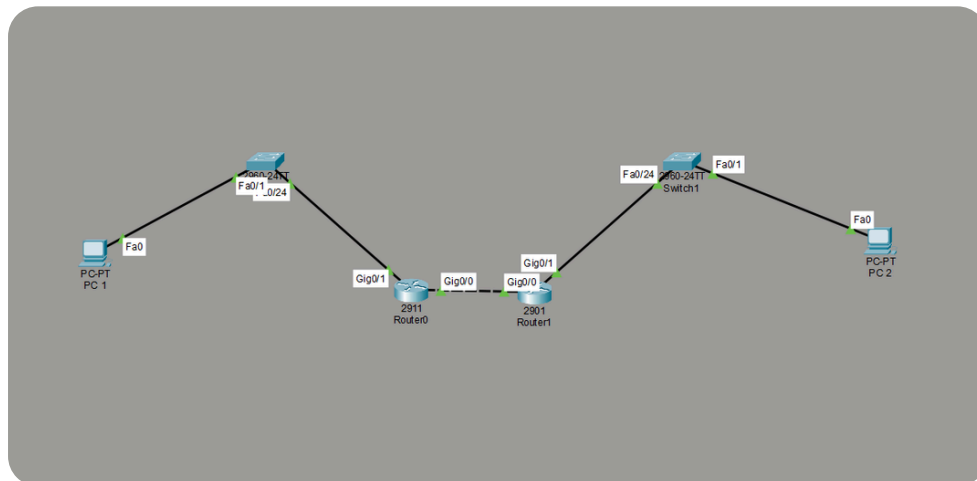
Internship Project Report

Project Repository: <https://github.com/SandeepSandy08/Cisco-Packet-Tracer-Networking-Labs.git>

BASIC NETWORK TROUBLESHOOTING USING PING AND TRACEROUTE

To understand how to diagnose, isolate, and resolve network connectivity issues across a routed infrastructure using standard diagnostic commands like ping and traceroute.

NETWORK TOPOLOGY:



DEVICES USED:

1. 2960 Switch - 2
2. 2911 Router - 1
3. 2901 Router - 1
4. PC(s) - 2
5. Cable type - Straight-Through Cable, Cable Cross-Over

IP ADDRESSING :

Devices	Interface	IP Address	Subnet Mask
PC 1	Fa0 → Fa0/1	192.168.11.8	255.255.255.0
Switch 1	Fa0/24 → Gig0/1	NIL	NIL
Router 1	Gig0/1 (LAN) Gig0/0 (WAN)	192.168.11.1 10.0.0.1	255.255.255.0
Router 2	Gig0/0 (WAN) Gig0/1 (LAN)	10.0.0.2 192.168.13.1	255.255.255.0
Switch 2	Fa0/24 → Gig0/1	NIL	NIL
PC 2	Fa0 → Fa0/1	192.168.13.3	255.255.255.0

CISCO PACKET TRACER NETWORK LABS

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CONFIGURATION :

- PCs - Configured with fixed static IP addresses and default gateways pointing to their local router interfaces.
- Routers - Configured with static network paths (ip route) pointing across the point-to-point WAN crossover link so both networks can find each other.
- The GitHub Repository Contains :
 1. Configuration files: Available in the GitHub repository under the Config Files folder.
 2. Packet Tracer lab file: Available under the Labs folder.

VERIFICATION & TROUBLESHOOTING

In Router CLI (type) - show ip route and show ip int br

To Verify

- Check if both intermediate router interfaces are up and active.
- Verify that the static routes are loaded in the routing table (marked with an S).

In PC Cmd (type) - tracert 192.168.13.3 (Ip address)

To Troubleshoot & Fix

- If the traceroute stops at the local gateway (192.168.11.1), it proves the local link is fine, but the router is missing a path to the destination.
- Add the missing ip route commands to the router configuration via the GitHub repository files to bridge the communication gap.

RESULT

- Pings and traceroutes between PC 1 and PC 2 now succeed. The traceroute command successfully displays every intermediate router interface hop along the path, proving that the network configuration errors were resolved and end-to-end routing is fully functional.

```
C:\>tracert 192.168.13.3

Tracing route to 192.168.13.3 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.11.1
  2  1 ms      0 ms      0 ms      10.0.0.2
  3  0 ms      0 ms      0 ms      192.168.13.3

Trace complete.
```

Traceroute Output

```
Router>
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet0/1
L       192.168.11.1/32 is directly connected, GigabitEthernet0/1
S       192.168.13.0/24 [1/0] via 10.0.0.2
```

Routing Table

CISCO PACKET TRACER NETWORK LABS

Internship Project Report

Project Repository: <https://github.com/SandeepSandy08/Cisco-Packet-Tracer-Networking-Labs.git>

CONCLUSION

- This internship gave me great hands-on practice with real-world networking concepts inside Cisco Packet Tracer. Building these labs helped me understand how traffic moves across a network, starting with simple switch setups and VLAN segregation, all the way up to advanced configurations like routing, DHCP servers, security, and DNS name resolution.
- Working through the troubleshooting tasks also taught me how to think like a network engineer—using everyday commands like ping and traceroute to find and fix broken links quickly. Finally, saving my network files and device scripts in a clean GitHub repository made it easy to keep my work organized and practice proper documentation habits.